A Dynamical Attractor Metric for the Conscious Subject

A Testable Operational Framework Based on Neural State-Space Invariants

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Abstract

Understanding the physical basis of the conscious subject remains an open problem in neuroscience. Existing frameworks describe mechanisms associated with conscious processing but rarely define an operational metric for the presence of a subject.

This work proposes a minimal dynamical hypothesis: conscious states may correlate with the presence of a stable reflexive attractor in neural state space.

From this hypothesis we derive a measurable scalar metric, Σ, constructed from experimentally estimable dynamical invariants:

- predictive integration (Ψ)
- dynamical complexity (C)
- predictive stability (F)

We present a computational pipeline allowing Σ to be estimated from EEG or MEG recordings using state-space reconstruction techniques.

The model generates a falsifiable prediction:

$$\boldsymbol{\Sigma_{awake}} > \boldsymbol{\Sigma_{sleep}} > \boldsymbol{\Sigma_{anesthesia}}$$

If confirmed, this would support the hypothesis that conscious states are associated with identifiable dynamical attractor structures in neural activity. If not, the attractor hypothesis is weakened.

1. Introduction

The nature of the conscious subject remains debated across neuroscience, philosophy, and cognitive science. Most contemporary frameworks explain aspects of conscious processing but do not provide a direct operational measure of the subject as a dynamical entity.

Several theoretical developments motivate a dynamical perspective:

- large-scale brain activity exhibits attractor-like dynamics
- predictive processing suggests internal self-modeling loops
- conscious states correlate with structured neural complexity

These observations motivate the following hypothesis.

2. Hypothesis

Dynamical Subject Hypothesis

Conscious states are expected to correlate with the presence of a stable reflexive attractor within the neural state space of a cognitive system.

Operational prediction:

Systems instantiating such an attractor will exhibit measurable dynamical invariants distinguishing conscious from unconscious states.

3. Neural State Space

Let the global neural state be:

$$\boldsymbol{X(t) = (x_1, x_2, \dots, x_n)}$$

where each variable represents a functional neural signal (EEG/MEG channels or derived components).

The system evolves as:

$$\dot{X} = F(X, U, \eta)$$

with:

U = sensory input

η = physiological noise

State space reconstruction uses delay embedding:

$$X_t = (x_t, x_{t-\tau}, x_{t-2\tau}, \dots)$$

(Takens embedding).

4. Operational Metric of the Subject

We define a scalar indicator:

$$\Sigma = \Psi \cdot C \cdot e^{-\beta F}$$

where each term is experimentally estimable.

4.1 Predictive Integration (Ψ)

Predictive integration measures how accurately the system predicts its own state.

Let

$$\widehat{X_{t+1}} = f(X_t)$$

be a local predictive model trained on neural time series.

Define prediction error:

$$E_t = \left\| X_{t+1} - \widehat{X_{t+1}} \right\|$$

Then

$\backslash Psi$ =

$$\frac{1}{1 + \langle E_t \rangle}$$

Higher Ψ indicates stronger internal predictive coherence.

4.2 Dynamical Complexity (C)

Complexity is estimated using attractor dimension.

Using the Grassberger–Procaccia algorithm:

C = D

where D is the correlation dimension of the reconstructed attractor.

Higher values indicate richer dynamical structure.

4.3 Predictive Free Energy (F)

Predictive mismatch term.

$$\boldsymbol{F} = \langle \boldsymbol{E_t}\text{^}\mathbf{2} \rangle$$

Lower values indicate better predictive stability.

5. Experimental Prediction

If the dynamical subject hypothesis is correct:

$$\boldsymbol{\Sigma_{awake}} > \boldsymbol{\Sigma_{REM}} > \boldsymbol{\Sigma_{deep\ leep}} > \boldsymbol{\Sigma_{anesthesia}}$$

These differences should be statistically detectable across subjects.

6. Experimental Pipeline

Step 1 — Data acquisition

EEG or MEG recordings

Suggested datasets:

sleep EEG datasets

anesthesia recordings

resting-state EEG

Step 2 — Preprocessing

Bandpass filter:

0.5–120 Hz

Artifact removal:

ICA decomposition

Normalization:

z-score across channels

Step 3 — State space reconstruction

Estimate embedding parameters:

$\tau \rightarrow$ mutual information minimum

$m \rightarrow$ false nearest neighbors

Construct delay embedding.

Step 4 — Attractor analysis

Compute:

correlation dimension (Grassberger–Procaccia)

Lyapunov exponent

entropy measures

Step 5 — Predictive modeling

Train a local predictor:

examples:

autoregressive model

reservoir computing

nonlinear regression

Compute prediction error.

Step 6 — Compute Σ

$$\boldsymbol{\Sigma = \Psi C e^{-\beta F}}$$

with β set empirically to normalize scale.

7. Statistical Analysis

Compare Σ across states using:

ANOVA

pairwise t-tests

Significance threshold:

$p < 0.05$

Correlation with clinical consciousness scores may also be tested.

8. Falsification Criteria

The dynamical subject hypothesis is weakened if:

1. Σ does not distinguish conscious from unconscious states.

2. reconstructed neural dynamics show no stable attractor structure.

3. predictive integration does not correlate with consciousness level.

9. Simulation Validation

Before empirical testing, the framework can be validated using simulated neural networks:

$$\frac{dx_i}{dt} = -x_i + \sum_j W_{ij} \tanh(x_j) + \eta_i(t)$$

Networks implementing internal predictive loops should exhibit higher Σ.

10. Discussion

The proposed metric provides an operational framework for studying the conscious subject as a dynamical phenomenon.

Unlike purely informational or computational approaches, this model focuses on state-space invariants of neural dynamics.

If supported empirically, this would suggest that subjective identity corresponds to a stable reflexive attractor within neural dynamics.

Importantly, the proposed framework does not assume that dynamical complexity alone implies the presence of a conscious subject. Complex attractor dynamics can arise in many physical systems without subjective properties.

The present hypothesis instead suggests that conscious states may correspond specifically to reflexive attractor dynamics, in which the system exhibits internally coherent predictive modeling of its own state.

An additional possibility is that the proposed metric may relate to the hypothesis that cortical dynamics operate near critical regimes between order and chaos. Systems near such critical points are known to exhibit high dynamical complexity while retaining global stability.

If so, the Σ metric might reach maximal values in neural regimes approaching criticality, consistent with empirical observations linking conscious states to complex but structured brain activity.

Further work would be required to evaluate this relationship directly.

11. Conclusion

This work introduces a falsifiable hypothesis: conscious states may be associated with reflexive dynamical attractors measurable through neural state-space invariants.

The metric Σ provides a practical method for estimating this property from EEG or MEG data.

Future studies can test this hypothesis using publicly available datasets and independent laboratories.

Metric refinement (post-publication note)

After the initial formulation of the Σ metric,

a stability correction term was introduced:

$$\Sigma^* = \Psi \cdot C \cdot \exp(-\beta F) \cdot \exp(-\lambda)$$

where λ approximates the largest Lyapunov exponent of the reconstructed attractor.

This correction penalizes unstable or chaotic dynamics and improves discrimination between stochastic neural activity and coherent attractor dynamics.

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